

Cem Özçelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with 4+ years building production ML systems - demand forecasting, anomaly detection, and end-to-end pipeline deployment. Industrial Engineering background means the first question is always "what does it cost when this model is wrong?" - not "what is the accuracy?" Built an unsupervised anomaly detection framework at Birinci Automotive (15% OEE gain, 15% increase in critical machine availability). Deployed ML supply chain forecasting at Anadolu that sustained an 8-hour weekly reduction in operational overtime. Demand forecasting portfolio: 12.83% RMSPE across 1,115 Rossmann stores with time-based validation and FastAPI inference. Experienced across the full lifecycle: Python, XGBoost, PySpark, Docker, FastAPI - from distributed ETL to containerized production APIs. Case studies: [Rossmann Forecasting](#) | [Credit Risk MLOps](#) | [Spark Rec Engine](#)

CORE COMPETENCIES

Demand Forecasting & Time-Series	Anomaly Detection Systems	ML Pipeline Architecture	XGBoost / Gradient Boosting
PySpark / Distributed Computing	FastAPI & Docker Deployment	Cost-Function Optimization	A/B Testing & Experiment Design

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 - Present

Senior Database Analyst | Maltepe, Istanbul

- Architected and deployed ML demand forecasting pipelines, sustaining an **8-hour weekly reduction in operational overtime** through Python-based supply chain algorithms and AI-driven prediction.
- Scaled a custom Django web application into a centralized hub for deploying multiple ML models via RESTful APIs - enabling rapid iteration and deployment of new prediction models.
- Led cross-functional data initiatives, translating SKU-level Association Rule Mining outputs into strategic business actions driving continuous revenue growth.
- Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker

Anadolu Pet Akvaryum & Effeffe

Sept 2024 - Dec 2025

IT & Data Analytics Specialist | Maltepe, Istanbul

- Engineered initial predictive supply chain optimization models using Python and PL/SQL, **reducing stockouts and improving inventory turnover**.
- Conducted Route-Level Profitability Analysis with advanced SQL, defining cost-to-profit ratios that delivered a **3% improvement in overall logistics efficiency**.
- Architected and deployed a full-stack internal web application from scratch using Python (Django) and SOLID principles - establishing foundational MLOps deployment practices.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 - Sept 2024

Mid-Level Data Analyst | Cayirova, Kocaeli

- Developed a **Predictive Maintenance & Anomaly Detection framework** using unsupervised clustering on historical machine failure patterns, delivering a **15% weekly increase in critical machine availability** and **15% OEE improvement**.
- Processed and analyzed high-frequency time-series sensor data from machine sensors for cycle time optimization - direct foundation for sequential demand modeling.
- Engineered robust ETL pipelines in MSSQL and Python to ingest high-volume manufacturing and sensor data, replacing manual reporting workflows.
- Tech:** Python, MSSQL, ETL, unsupervised clustering, time-series analysis

Junior Data Analyst | Cayirova, Kocaeli

- Conducted EDA and applied statistical methods to workforce datasets, uncovering patterns to optimize HR policies.
- Designed and implemented automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

Retail Demand Forecasting - End-to-End MLOps Pipeline

Forecasting

~12.83% **RMSPE** on 6-week daily demand forecasting across 1,115 Rossmann stores using time-based validation (no temporal leakage). OOP training pipeline (DataIngestion → DataTransformation → ModelTrainer). FastAPI REST inference + Streamlit dashboard with **dynamic confidence intervals** translating ML metrics to business terms. Surfaced that B-type stores yield lower promotion ROI vs A/C - enabled marketing budget reallocation.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | github.com/ccemozclk/Rossmann-Demand-Forecasting

End-to-End Credit Risk Prediction - MLOps & Anomaly Detection

Risk / Fraud

63% F1-Score with XGBoost on Lending Club dataset, optimized for heavy class imbalance with temporal holdout validation (no leakage). Full MLOps pipeline: raw ingestion → feature engineering → containerized production app. Architecture pattern transfers directly to **fraud detection, payment intelligence, and behavioral anomaly modeling**. Interactive EDA platform (Django) for non-technical reviewers.

Python, XGBoost, CatBoost, Django, Docker, Scikit-learn | github.com/ccemozclk/Credit-Risk

Scalable Recommendation Engine - PySpark & Customer Segmentation

Distributed ML

End-to-end personalized recommendation system on Apache Spark/Databricks. **RFM + K-Means** behavioral segmentation into 5 strategic groups; **ALS collaborative filtering** with implicit feedback for item-level recommendations. Cold-start handling built in. Segmentation logic directly applicable to driver/rider behavioral modeling and dynamic supply-demand matching.

PySpark, Databricks, K-Means, ALS, Streamlit | github.com/ccemozclk/Spark-Recommendation-Engine

EDUCATION

B.Sc. Industrial Engineering - Kirikkale University

2016 - 2021

MILP optimization (IBM ILOG CPLEX), statistical analysis, multi-criteria decision making (AHP, TOPSIS). Foundation for cost-function thinking in ML system design.

Associate Data Scientist Career Track - DataCamp

Jan - May 2024

PySpark, Feature Engineering, Recommendation Engines (ALS). Distributed computing track.

Data Science & Machine Learning Bootcamp - Miuul

Mar - Sept 2022

6-month intensive covering the end-to-end ML lifecycle. 10+ real-world projects in classification, regression, and clustering.

CERTIFICATIONS

Associate Data Scientist - DataCamp

2024

SKILLS

ML / DL: XGBoost, LightGBM, Scikit-learn, TensorFlow, Keras, PyTorch, CNNs, RNNs, Transfer Learning

MLOps: Docker, FastAPI, Django, REST APIs, Git, Linux, End-to-End Deployment

Big Data: PySpark, Databricks, Apache Spark, AWS (EC2, S3), Distributed Computing

Data: Python (Advanced), SQL (Advanced), PostgreSQL, MSSQL, Oracle, ETL Pipelines, Pandas, NumPy

Analytics: A/B Testing, Experiment Design, Cost-Function Optimization, MILP, AHP/TOPSIS **Languages:** Turkish (Native), English (Fluent)